

Investigating the Effect of Central Thalamic Deep Brain Stimulation on Sleep in Alzheimer’s Disease Model

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Synopsis

Motivation: In Alzheimer’s disease (AD), the neuropsychiatric inventory is strongly affected by sleep disorders, and vice versa. Central thalamic deep brain stimulation (CT-iTBS) has improved memory and altered the hypothalamic function which may impact the orexinergic system and sleep.

Goal(s): Investigating the therapeutic effect of CT-iTBS on sleep, orexinergic system, and cognitive function in AD.

Approach: Applied functional magnetic resonance imaging, electrocorticogram-electromyogram, behavioral tasks, and ELISA to assess the impact of CT-iTBS in 3xTgAD mouse model.

Results: CT-iTBS significantly improved sleep fragmentation, functional connectivity, cognitive function, and orexin receptors and concentrations in 3xTgAD mice.

Impact: We discovered that CT-iTBS may play an important role in modulating sleep, the orexinergic system, and cognitive function in AD. Improved outcomes pave the future direction of treating sleep disorders in Alzheimer's disease.

Introduction

The prevalence of sleep disorders in neurodegenerative diseases is high, especially in Parkinson’s disease and Alzheimer’s disease (AD)^{1,2}. Previous studies have shown multiple types of sleep disorders were associated with Alzheimer’s disease and progressed as the disease worsened³. The orexinergic system had been found to modulate and regular wake-sleep cycle⁴⁻⁶. Lots of evidence shows that the orexinergic system had an important role in activating and modulating memory and motivated behavior function^{7,8}. Recently, intermittent theta-burst stimulation (iTBS) has been hypothesized to alter cognitive functions in a more efficacious way. By delivering in rhythmic bouts of frequency at 3-8 Hz, iTBS recapitulates natural brain rhythm to immediately provoke theta oscillations in the brain, which have profound functional relevance to cognition⁹. The central thalamus (CT), a signal hub in the brain, had been considered to be the target for enhancing cognitive functions¹⁰. In this study, we used functional magnetic resonance imaging (fMRI), electrocorticogram-electromyogram (ECoG/EMG), behavioral tasks, and orexin to evaluate the therapeutic effect of CT-iTBS in 3xTgAD mice on sleep.

Methods

8-month-old male C57BL/6 mice (weight 20 ± 5 g, *N* = 14) were used as healthy controls, and age-match male 3xTgAD mice (weight 20 ± 5 g, *N* = 28) were used as sham controls and DBS group. All mice received implantation surgery with two MR-compatible neural probes into the bilateral CT (AP: – 1.56 mm, ML: ± 0.7 mm, DV: – 3.0 mm). Also, half of mice in the three groups placed screw electrodes over the skull to record the ECoG signals. After seven days of recovery, implanted mice in three groups were placed in a plastic cage (30 cm diameter and 38 cm height) for 30 min/day with/without iTBS (Biphasic electrical current of 150 µA with a pulse width of 100 µs and burst width of 100 ms at 5 Hz burst rate and 200 Hz intra-burst rate (10 pulses per burst)) and then performed behavioral tasks. Open field task (OFT) and novel object recognition (NOR) over three days to evaluate anxiety and cognitive function. To investigate functional connectivity (FC) alteration with CT-iTBS, brain images were obtained using a Bruker 7 Tesla MRI scanner (Bruker Biospec 70/30 USR, Ettlingen, Germany). The gradient-echo planar imaging (GE-EPI) sequence (TR = 2,000 ms, TE = 20 ms, FOV = 20 × 20 mm², matrix size = 80 × 80, bandwidth = 200 kHz, 14 coronal slices, and thickness = 0.5 mm) was used to obtain the resting-state fMRI (rsfMRI) images. There were 260 dynamics for 10 dummy scans and 250 rsfMRI images in the GE-EPI images. The FMRIB Software Library v5.0 (FSL 5.0; www.fmrib.ox.ac.uk/fsl) and the Analysis of Functional NeuroImages (AFNI) software (<http://afni.nimh.nih.gov/afni>) were used to perform a ROI-based analysis of FC between regions such as the prefrontal cortex (PFC), CT, hippocampus (HIP), and hypothalamus (HYPO). The Allen mouse brain¹¹ ROIs were also resliced to match the affined GE-EPI datasets in dimension. To compare the FC caused by CT-iTBS quantitatively, the measured FC between distinct brain areas was normalized. Then, sacrificed and randomly divided equally for IHC staining (Anti-Orexin receptor 1 (OXR1, AOR-001, Alomone)) and ELISA (Mouse Orexin A/Hypocretin-1 ELISA Kit (NBP2-80231, Novus-Biologicals)).

Results

In behavioral tasks, DBS group showed significantly higher PI, total distance, and time spent in the inner zone compared with sham controls (**Fig. 1**). The rsfMRI results revealed a significant increase in functional connectivity (FC) between PFC, CT, HIP, and HYPO following CT-iTBS treatment (**Fig. 2**). Meanwhile, we measured fragmentation index (FI), calculated as total number of awakenings divided by total sleep time¹². The FI in dark phase showed significantly higher in sham controls compared with healthy controls and DBS group (**Fig. 3**). Additionally, the concentration of orexin-1 in HYPO was significantly lower in sham controls and increased after CT-iTBS treatment (**Fig. 4**).

Discussion

We found CT-iTBS improved the performance of behavioral tasks related to anxiety and cognitive function, enhanced FC, decreased FI, and enhanced concentration of orexin-1 in HYPO. In previous studies, the DBS mechanism has focused on manipulating brain networks^{10,13-15}. CT-iTBS was associated with corticolimbic and thalamocortical networks, which are related to modulate consciousness, sleep/wake cycle, and cognitive function^{10,16}. Orexin neuron loss and lower concentration were found in AD patients¹⁷ related to sleep and memory^{7,18}.

Conclusion

In this study, we applied the CT-iTBS in the 3xTgAD mice model, demonstrating the differences in behavioral performance, FC, sleep fragmentation, and orexin in healthy controls, sham controls, and DBS group. Our findings support the importance of CT as a therapeutic target for improving sleep, orexinergic system and cognitive function in AD individuals.

Acknowledgements

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Figures

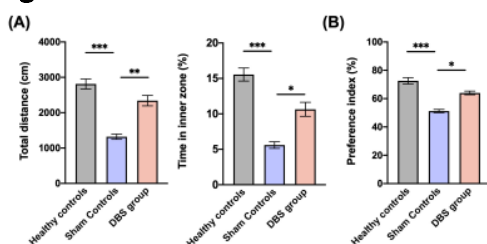


Figure 1. (A) The total distance and percentage of time in inner zone showed significant decrease in sham controls compared with healthy controls and DBS group. (B) Preference index was significantly lower in sham controls than in healthy controls and DBS group.

*, **, and *** indicated significant differences with $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively, and the data were analyzed using the Kruskal-Wallis test (Mean $\pm$ SEM).

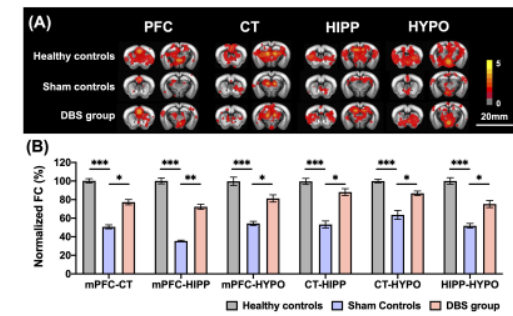


Figure 2. (A) The group comparisons of cross-correlation maps on PFC, CT, HIPP, HYPO for healthy controls, sham controls, and DBS group. (B) The percentage of normalized FC between ROIs to all the other ROIs after CT-iTBS in AD mice. The DBS group showed a widespread enhancement of FC compared with sham controls. *, **, and *** indicated significant differences with $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively, and the data were analyzed using the Kruskal-Wallis test (Mean $\pm$ SEM).

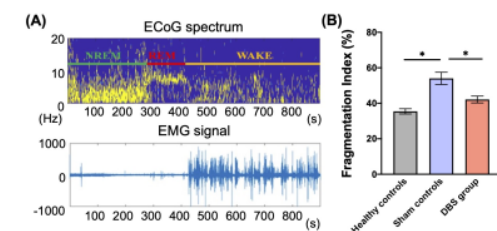


Figure 3. (A) The representative session of the ECoG spectrum and EMG signal were recorded to identify the timing of wake, NREM, and REM sleep states. (B) The fragmentation index in the dark phase showed a significant increase in sham controls compared with healthy controls and DBS group.

* indicated significant differences with $p < 0.05$, and the data were analyzed using the Kruskal-Wallis test (Mean $\pm$ SEM).

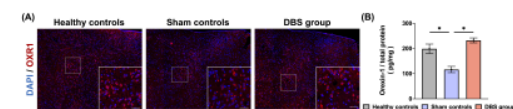


Figure 4. (A) Immunostaining of orexin receptor 1 antibody (OX1) on HYPO in three groups. Scale bar: 200 μ m; inset 40 μ m. (B) In DBS group, the concentration of orexin 1 in HYPO was significantly higher than in sham controls.

* indicated significant differences with $p < 0.05$, and the data were analyzed using the Kruskal-Wallis test (Mean $\pm$ SEM).